ROAD INSPECTOR

ROAD QUALITY INSPECTION
&
REPAIR ESTIMATION SYSTEM

Project ID:-R26-IT-023

GROUP MEMBERS



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AI & IoT-Based Road Defect Detection System



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WALALLAWITA K.L.T.D

AI-Based Repair Area Estimation using Semantic Segmentation



IT22113122

BANDARA T.C.N

Automated Material Requirement Estimation for Road Repairs



IT22207968

MANATHUNGA M.A.D.V.G

Environmental and Traffic Analysis with Repair Scheduling and Alternate Route Recommendation




ROAD INSPECTOR

INTRODUCTION



Road defects such as potholes and cracks are major problems that affect transportation safety and road quality.

Traditional road inspection methods are time-consuming, costly, and less efficient. This research project proposes an AI & IoT-Based Road Defect Detection System called Road Inspector to automate road monitoring and defect identification.

The system uses Artificial Intelligence, Computer Vision, IoT, and Environmental & Traffic Analysis to detect road damages, estimate repair areas, schedule repair activities, and recommend alternate routes during maintenance operations.

The proposed solution improves road maintenance efficiency, reduces human effort, and supports smarter and safer transportation infrastructure management.



USERS AND STAKEHOLDERS

Road Maintenance Planners

- Plan repair operations.

Municipal Road Authorities

- Monitor road conditions and manage maintenance budgets.

Maintenance Field Teams

- Perform road repair activities.

Material Estimation Engineers

- Calculate required construction materials.

Road Users

- Drivers and pedestrians affected by unsafe roads

IMPACT OF THE PROBLEM



1. SOCIAL IMPACT

- Increased risk of accidents
- Unsafe travel conditions

2. ECONOMIC IMPACT

- Vehicle damage
- Higher maintenance costs

3. TRANSPORTATION IMPACT

- Traffic congestion
- Travel delays

4. ENVIRONMENTAL IMPACT

- Waste of construction materials
- Increased emissions due to inefficient repairs

SYSTEM COMPONENTS

The proposed system consists of four intelligent components:

1. Road Defect Detection Module

2. Repair Area Estimation Module

3. Material Requirement Calculation Module

4. Environmental and Traffic Analysis with Repair Scheduling and Alternate Route Recommendation

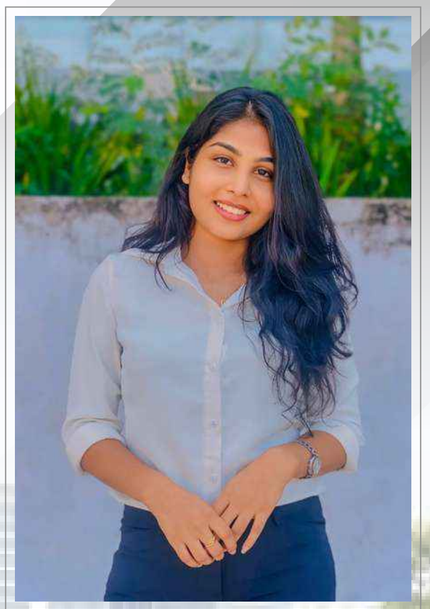








AI & IoT-Based Road Defect Detection System



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B.Sc. (Hons) Degree in Information Technology Specialization

in Information Technology

AI & IoT-Based Road Defect Detection System

How the Solution Addresses the Sub-Problem



Image Upload & Road Data Input
Road surface images are uploaded or captured through IoT-based camera devices for defect analysis.



AI-Based Defect Detection Engine
The system analyzes road images using AI and computer vision to detect potholes, cracks, and damaged road areas.



Detection Result Dashboard
Detected defects are displayed with defect type, severity level, confidence score, and visual output for further decision-making.

User Requirements Addressed



Real-Time Defect Identification
Supports fast detection of road surface damages without depending fully on manual inspection.



Improved Inspection Accuracy
AI-based analysis helps reduce human error and provides consistent road condition evaluation.



Support for Next Modules
Detection results can be shared with repair area estimation, material estimation, and repair scheduling modules.

AI & IoT-Based Road Defect Detection System

Design Excellence / Contribution



AI-Powered Defect Detection Framework
Integrates Artificial Intelligence, Computer Vision, and IoT technologies to automatically identify potholes, cracks, and road surface damages with high accuracy.



Real-Time Image Processing & Analysis
Processes uploaded or IoT-captured road images instantly to provide faster and more efficient road condition monitoring.

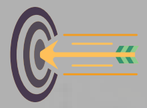


Integrated Detection & Severity Classification
Detected defects are displayed with defect type, severity level, confidence score, and visual output for further decision-making.

User Feedback on Prototype



Real-Time Defect Identification
Supports fast detection of road surface damages without depending fully on manual inspection.



Improved Inspection Accuracy
AI-based analysis helps reduce human error and provides consistent road condition evaluation.



Support for Next Modules
Detection results can be shared with repair area estimation, material estimation, and repair scheduling modules.

AI-Based Repair Area Estimation using Semantic Segmentation



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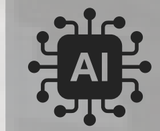
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AI-Based Repair Area Estimation using Semantic Segmentation

How the Solution Addresses the Sub-Problem



Road Image Input & Preprocessing
Road surface images are uploaded and preprocessed to improve image quality for accurate damage analysis.



Semantic Segmentation Engine
The system uses AI-based semantic segmentation techniques to identify and isolate damaged road regions at the pixel level.



Repair Area Estimation & Visualization Module
The system analyzes segmented damaged regions to calculate the exact repair area and visually displays the segmented outputs and repair measurements for maintenance planning and decision-making.

User Requirements Addressed



Accurate Repair Area Identification
Users observed that the system accurately highlights damaged road regions for repair planning.



Easy-to-Understand Visual Outputs
Segmented outputs clearly display damaged areas, making results simple to interpret.



Faster Repair Estimation Process
Automated area estimation reduces the time required for manual road damage measurements.

AI-Based Repair Area Estimation using Semantic Segmentation

Design Excellence / Contribution



AI-Based Semantic Segmentation Framework
Uses deep learning semantic segmentation techniques to identify damaged road regions at the pixel level with high precision.



Accurate Repair Area Calculation
Provides precise repair area measurements to support efficient road maintenance planning.

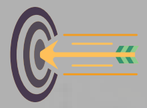


Reduced Manual Measurement Effort
Minimizes traditional manual inspection and measurement processes through intelligent automation.

User Feedback on Prototype



Real-Time Defect Identification
Supports fast detection of road surface damages without depending fully on manual inspection.



Improved Inspection Accuracy
AI-based analysis helps reduce human error and provides consistent road condition evaluation.



Support for Next Modules
Detection results can be shared with repair area estimation, material estimation, and repair scheduling modules.

Automated Material Requirement Estimation for Road Repairs



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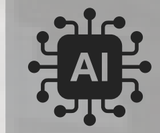
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Automated Material Requirement Estimation for Road Repairs

How the Solution Addresses the Sub-Problem



Repair Data Input & Analysis
The system receives repair area measurements and road defect data from previous modules for material estimation.



AI-Based Material Estimation Engine
AI algorithms analyze the damaged area and estimate the quantity of materials required for road repair operations.



Heat Requirement & Result Visualization Module
The system calculates the required heat levels for asphalt repair and displays estimated material quantities, heat requirements, and repair details for maintenance planning and decision-making.

User Requirements Addressed



Accurate Material Estimation
Provides precise material quantity calculations for efficient road repair operations.



Efficient Heat Requirement Analysis
Calculates suitable heat levels required for asphalt and road maintenance processes.



Reduced Resource Wastage
Helps minimize unnecessary material usage and repair costs through accurate estimation.

Automated Material Requirement Estimation for Road Repairs

Design Excellence / Contribution



AI-Based Material Estimation Framework
Uses intelligent algorithms to estimate required road repair materials accurately based on damaged area data.



Integrated Heat Requirement Analysis
Calculates optimal heat levels required for efficient asphalt repair and maintenance operations.

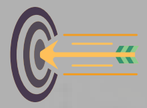


Automated Resource Optimization
Helps reduce material wastage and improve repair efficiency through smart estimation techniques.

User Feedback on Prototype



Accurate Material Estimation Results
Users observed reliable material quantity calculations for road repair operations.



Reduced Manual Calculation Effort
Automated estimation minimized the need for traditional manual material calculations.



Easy-to-Understand Repair Outputs
Estimated material quantities and heat requirement details were clearly displayed and easy to analyze.

Environmental and Traffic Analysis with Repair Scheduling and Alternate Route Recommendation



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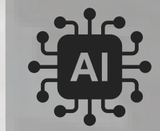
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Automated Material Requirement Estimation for Road Repairs

How the Solution Addresses the Sub-Problem



Environmental & Traffic Data Collection
The system collects real-time environmental and traffic data from sensors, cameras, and external data sources.



Traffic & Environmental Analysis Engine
AI algorithms analyze weather conditions, traffic density, and road usage patterns to support maintenance planning.



Smart Repair Scheduling & Alternate Route Recommendation Module
The system analyzes traffic flow and environmental conditions to recommend optimal repair schedules and generate alternate routes that minimize traffic congestion and improve road user safety during maintenance operations.

User Requirements Addressed



Efficient Repair Scheduling
Recommends suitable repair times to reduce traffic disruptions during maintenance activities.



Improved Traffic Management
Provides alternate route recommendations to minimize congestion and improve traffic flow.



Real-Time Environmental Analysis
Analyzes environmental conditions to support safer and more effective road repair planning.

Automated Material Requirement Estimation for Road Repairs

Design Excellence / Contribution



AI-Based Traffic & Environmental Analysis
Uses intelligent algorithms to analyze traffic patterns and environmental conditions in real time.



Smart Repair Scheduling System
Generates optimized repair schedules to minimize traffic disruptions and maintenance delays.

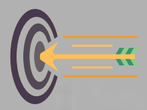


Intelligent Alternate Route Recommendation
Provides safer and less congested alternate routes during road repair operations.

User Feedback on Prototype



Effective Traffic Management Support
Users observed that alternate route recommendations helped reduce traffic congestion during maintenance activities.



Improved Repair Planning Efficiency
Smart repair scheduling provided better coordination and minimized unnecessary road disruptions.



Enhanced Road User Safety
The system helped improve safety by recommending safer travel routes during road repair operations.



ABILITY OF COMMERCIALIZATION

Target Market:
Road development authorities, municipal councils, highway maintenance departments, and smart city infrastructure projects.

Value:
AI-driven road defect detection, repair area estimation, material estimation, and smart maintenance planning to improve road maintenance efficiency and reduce operational costs.

Revenue Model:
Subscription-based platform services, government infrastructure solutions, maintenance analytics services, and smart road monitoring solutions.

Advantage:
Integrated AI, IoT, semantic segmentation, traffic analysis, and material estimation in a single intelligent road maintenance platform.

Market Potential:
Supports the growing demand for smart transportation infrastructure and automated road maintenance systems in modern cities and developing countries.


THANK YOU.